

Power Budget

Team Number:	308
Project Name:	
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Version:	

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators,

All Major Components	Component Name	Part Number	Voltage Range	#	Maximum Current	Current(mA)	Unit
	ESP32 Module	ESP32-WROOM-	3.0V - 3.6V	1	1000	1000	mA
	Photoresistor	GL5528	3.3V - 5V	1	Passive	0	mA
						0	mA
						0	mA
						0	mA
						0	mA

B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in

+12V Power Rail							
	Component Name	Part Number	Voltage Range	Quantity	Maximum Current	Allocated Current(mA)	Unit
c1. Regulator or Source Chip	Solenoid	(full part number)	+12 - 24V	1	500	500	mA
	Stepper motor	(full part number)	+12 - 24V	1	300	300	mA
						0	mA
						0	mA
						0	mA
						800	mA
						25%	
						1000	mA
Total Current Required on +12V Rail							
Safety Margin							
Subtotal							
Total Remaining Current Available on +12V Rail							
+5V Power Rail							
	Component Name	Part Number	Voltage Range	Quantity	Maximum Current	Allocated Current(mA)	Unit
	PSoC™ 4 BLE Module	(full part number)	(range)	2	200	400	mA
	Opamp	(full part number)	(range)	1	100	100	mA

						0	mA
						0	mA
						0	mA
					<i>Subtotal</i>	500	mA
					<i>Safety Margin</i>	25%	
					<i>Total Current Required on +5V Rail</i>	625	mA
c2. Regulator or Source Channel	+5V Regulator	LM7805	(range)	1	1000	1000	mA
					<i>Total Remaining Current Available on +5V Rail</i>	375	mA
-5V Power Rail							
	Component Name	Part Number	Voltage Range	#	Maximum Current	Current(mA)	Unit
	Opamp	(full part number)	(range)	1	100	100	mA
						0	mA
						0	mA
						0	mA
					<i>Subtotal</i>	100	mA
					<i>Safety Margin</i>	25%	
					<i>Total Current Required on -5V Rail</i>	125	mA
c3. Regulator or Source Channel	-5V Regulator	(full part number)	(range)	1	500	500	mA
					<i>Total Remaining Current Available on -5V Rail</i>	375	mA
+3.3V Power Rail							
	Component Name	Part Number	Voltage Range	#	Maximum Current	Current(mA)	Unit
	ESP32 Module	ESP32-WROOM-	3.0V - 3.6V	1	1000	1000	mA
						0	mA
						0	mA
						0	mA
					<i>Subtotal</i>	1000	mA
					<i>Safety Margin</i>	25%	
					<i>Total Current Required on +3.3V Rail</i>	1250	mA
c4. Regulator or Source Channel	+3.3V low-dropout regulator	LM1085ISX-3.3	+5V - 15V	1	3000	3000	mA
					<i>Total Remaining Current Available on 3.3V Rail</i>	1750	mA

C. For each power rail above, select a specific voltage regulator using the same process as for major component

D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of

External Power Source 1		Component Name	Part Number	yVoltageRange	Output	Maximum Current	Current(mA)	Unit
Power Source 1 Selection		AC-DC Wall Supply	Model 0930	110-220VAC	+9V	3000	3000	mA
Power Rails Connected to External Power Source 1							0	mA
							0	mA
		+3.3V low-dropout regulator	LM1085ISX-3.3	+5V - 15V	1	3000	3000	mA
		Total Remaining Current Available on External Power Source 1					0	mA
External Power Source 2		Component Name	Part Number	yVoltageRange	Output	Maximum Current	Current(mA)	Unit
Power Source 2 Selection		Battery	full part number	+9V	-9V	500	500	mA
Power Rails Connected to External Power Source 2		-5V Regulator	(full part number)	(range)	1	500	500	mA
		Total Remaining Current Available on External Power Source 2					0	mA
		E. Calculate Battery Life (if applicable). For each battery, also check the worst-case lifetime of the						
		Component Name	Part Number	yVoltageRange	Capacity(mAh)	Used By Regulators		
		Battery	(full part number)	+12V	500	500		
		Battery Life					1	hours

Notes

External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).
If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit.